

Mass from Local Curvature Thresholds in a Scalar Topographic Effective Field Theory

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We develop a scalar topographic effective field theory (EFT) in which localized, static field configurations acquire mass through a geometric matching condition involving a local curvature scalar. The theory introduces a curvature operator $\mathcal{K}[\rho] = -\nabla^2 \ln \rho$ acting on a scalar density $\rho(x)$, and defines an effective local curvature k_{loc} evaluated at the core radius of a localized mode. A dimensionless activation number $N_{\mathcal{K}}$ normalizes the curvature content of the configuration, and the emergent mass functional

$$m = \frac{c^2}{2G} r_c^2 k_{\text{loc}}$$

is shown to be consistent with dimensional analysis, stability considerations, and matching to the Newtonian gravitational potential in the infrared. The EFT remains agnostic about microphysical origins while providing a universal geometric mechanism for mass generation. We analyze stability, EFT validity, profile independence, and curvature discontinuities, and we propose experimental tests based on electromagnetic curvature excitation in conductive shells. The framework is self-contained, falsifiable, and compatible with general EFT principles.

I. INTRODUCTION

The origin of mass remains one of the most fundamental questions in theoretical physics. While the Higgs mechanism successfully explains the masses of elementary particles within the Standard Model, and while gravitational mass is encoded in the stress-energy tensor in general relativity (GR), there remains conceptual space for alternative or complementary mechanisms in which mass emerges from geometric or topographic properties of field configurations.

In this work we develop a scalar topographic effective field theory (EFT) in which mass arises from a *local curvature threshold* associated with a scalar density profile. The theory introduces a curvature operator acting on the logarithm of the density,

$$\mathcal{K}[\rho](x) \equiv -\nabla^2 \ln \rho(x), \quad (1)$$

and defines an effective local curvature scalar k_{loc} by evaluating $\mathcal{K}$ at the core radius of a localized configuration. A dimensionless activation number $N_{\mathcal{K}}$ normalizes the curvature content of the configuration, ensuring that a single localized mode corresponds to a single curvature activation.

The central result of this framework is a mass functional of the form

$$m = \frac{c^2}{2G} r_c^2 k_{\text{loc}}, \quad (2)$$

which is not derived from the action but is instead a geometric matching ansatz motivated by (i) dimensional consistency, (ii) the requirement that mass be extensive in the dimensionless combination $r_c^2 k_{\text{loc}}$, and (iii) matching to the Newtonian gravitational potential in the infrared.

This paper develops the mathematical structure of the theory, analyzes its stability and EFT consistency, and proposes experimental tests based on electromagnetic curvature excitation in conductive shells. The framework is self-contained and does not rely on cosmological or microphysical assumptions, making it suitable as a standalone mid-scale EFT.

The remainder of the paper is organized as follows. Section II defines the curvature operator, activation number, and effective curvature scalar. Section III derives the mass functional and its normalization. Section IV discusses EFT validity and stability. Section V analyzes profile independence. Section VI proposes experimental tests. We conclude in Sec. VIII.

II. LOCAL CURVATURE OPERATOR AND CURVATURE ACTIVATION

In this section we define the curvature operator $\mathcal{K}[\rho]$, the dimensionless activation number $N_{\mathcal{K}}$, and the effective local curvature scalar k_{loc} that together form the geometric core of the scalar topographic EFT. These quantities are defined independently of the underlying microphysics and rely only on the scalar density profile $\rho(x)$ and the background metric $g_{\mu\nu}$.

A. Local curvature operator

Let (M, g) be a smooth, oriented, four-dimensional Lorentzian manifold with metric $g_{\mu\nu}$. Let $\rho(x)$ be a positive-definite scalar density associated with a static field configuration. We define the *local curvature operator* by

$$\mathcal{K}[\rho](x) \equiv -\nabla^2 \ln \rho(x), \quad (3)$$

where ∇^2 is the Laplace–Beltrami operator,

$$\nabla^2 f = \frac{1}{\sqrt{|g|}} \partial_\mu \left(\sqrt{|g|} g^{\mu\nu} \partial_\nu f \right). \quad (4)$$

The operator $\mathcal{K}$ has dimensions

$$[\mathcal{K}] = \text{length}^{-2}, \quad (5)$$

and is invariant under multiplicative rescalings of ρ , since $\ln \rho$ shifts only by a constant.

In a locally flat, spherically symmetric region with line element $ds^2 = -c^2 dt^2 + dr^2 + r^2 d\Omega^2$, Eq. (3) reduces to

$$\mathcal{K}[\rho](r) = -\frac{d^2}{dr^2} \ln \rho(r) - \frac{2}{r} \frac{d}{dr} \ln \rho(r). \quad (6)$$

This form makes explicit that $\mathcal{K}$ measures the *local curvature content* of the profile $\rho(r)$, independent of its overall normalization.

B. Core radius and effective curvature scalar

A localized configuration is characterized by a core radius r_c at which the radial gradient reaches an extremum:

$$\left. \frac{d}{dr} |\partial_r \rho(r)| \right|_{r=r_c} = 0. \quad (7)$$

We define the *effective local curvature scalar* by evaluating the curvature operator at the core radius:

$$k_{\text{loc}} \equiv \mathcal{K}[\rho](r_c). \quad (8)$$

This scalar quantity plays the central role in the mass functional introduced in Sec. III. It encodes the curvature activation of the localized mode and is independent of the overall normalization of ρ .

C. Dimensionless curvature activation number

To quantify the curvature content of a localized configuration, we define the *dimensionless curvature activation number*

$$N_{\mathcal{K}} \equiv \int_{r_c-\sigma}^{r_c+\sigma} \sigma \mathcal{K}[\rho](r) dr, \quad (9)$$

where σ is the characteristic width of the profile.

Since $\sigma \mathcal{K}$ is dimensionless, the integral is dimensionless. A *single activation* is defined by the normalization condition

$$N_{\mathcal{K}} = 1. \quad (10)$$

For smooth profiles with slowly varying curvature across the core, Eq. (10) implies the approximate relation

$$k_{\text{loc}} \sim \frac{1}{2\sigma^2}, \quad (11)$$

up to order-unity factors depending on the detailed shape of $\rho(r)$.

This normalization ensures that the mass functional in Sec. III assigns the same mass to all profiles with the same curvature activation, regardless of their detailed shape.

D. Interpretation

The operator $\mathcal{K}$ measures the curvature of the logarithmic density profile, and k_{loc} captures the curvature activation at the core of the localized mode. The activation number $N_{\mathcal{K}}$ counts the number of curvature quanta supported by the configuration.

These quantities are purely geometric and do not depend on the microphysical origin of ρ . They form the foundation of the mass functional developed in the next section.

III. MASS FUNCTIONAL AND GEOMETRIC MATCHING ANSATZ

The scalar topographic EFT does not derive the mass of a localized configuration directly from the action. Instead, mass emerges from a geometric matching condition that ties the local curvature activation of the scalar density profile to the infrared gravitational potential. This section develops the mass functional, its normalization, and its consistency with Newtonian gravity.

A. Komar mass and infrared matching

In a static, asymptotically flat spacetime with timelike Killing vector ξ^μ , the Komar mass is defined by

$$m_{\text{Komar}} = \frac{1}{4\pi G} \int_{S_\infty^2} \nabla^\mu \xi^\nu dS_{\mu\nu}. \quad (12)$$

For a compact, localized configuration, the gravitational potential at large radius must reduce to the Newtonian form

$$\phi_{\text{grav}}(r) \longrightarrow -\frac{Gm}{r}, \quad r \rightarrow \infty. \quad (13)$$

The scalar topographic EFT requires that the mass assigned to a localized mode reproduce this infrared behavior. This requirement motivates the geometric matching ansatz introduced below.

B. Geometric matching ansatz

The effective local curvature scalar k_{loc} defined in Eq. (8) measures the curvature activation of the scalar density profile at its core radius r_c . Motivated by dimensional analysis, geometric extensivity, and infrared matching, we postulate the mass functional

$$m[\rho] \equiv \frac{c^2}{2G} r_c^2 k_{\text{loc}}. \quad (14)$$

This relation is not derived from the action; it is a *geometric normalization choice* that ensures:

- (i) the mass is extensive in the dimensionless combination $r_c^2 k_{\text{loc}}$,
- (ii) the normalization matches the Newtonian potential in Eq. (13),
- (iii) the mass depends only on the curvature activation of the localized mode, not on the overall normalization of ρ .

The factor $c^2/2G$ is fixed by requiring consistency with the Newtonian limit for a single curvature activation ($N_{\mathcal{K}} = 1$).

C. Dimensional analysis

The dimensions of the quantities entering Eq. (14) are

$$\left[\frac{c^2}{2G} \right] = \frac{\text{kg}}{\text{m}}, \quad [r_c^2] = \text{m}^2, \quad [k_{\text{loc}}] = \text{m}^{-2}. \quad (15)$$

Thus,

$$[m] = \text{kg}, \quad (16)$$

as required.

Because $r_c^2 k_{\text{loc}}$ is dimensionless, the mass functional is invariant under profile deformations that preserve the curvature activation number N_K .

D. Interpretation of the mass functional

The mass functional (14) assigns mass to a localized configuration based on its curvature activation. The quantity k_{loc} acts as a curvature threshold: only profiles with $k_{\text{loc}} > 0$ correspond to stable, localized modes (see Sec. IV). The mass is proportional to the curvature content of the core region and scales with the square of the core radius.

This structure is reminiscent of geometric mass generation mechanisms in higher-derivative gravity and topological field theories, but the present framework remains agnostic about microphysical origins. The mass functional is defined purely by geometric matching and EFT consistency.

E. Infrared consistency check

To verify that Eq. (14) reproduces the Newtonian potential, consider a localized configuration with a single curvature activation ($N_K = 1$). The effective curvature scalar scales as $k_{\text{loc}} \sim 1/(2\sigma^2)$, and the core radius satisfies $r_c \sim \sigma$ up to order-unity factors.

Then

$$m \sim \frac{c^2}{2G}, \quad (17)$$

which is the mass scale required to reproduce the Newtonian potential in Eq. (13) for a single activation.

More generally, configurations with $N_K = n$ curvature activations satisfy

$$m_n = n m_1, \quad (18)$$

ensuring extensivity and infrared consistency.

This completes the construction of the mass functional.

IV. EFT VALIDITY, STABILITY, AND CURVATURE DISCONTINUITIES

The scalar topographic EFT is intended as a low-energy, long-wavelength description of localized curvature activations. In this section we analyze the regime of validity, the stability of localized modes, and the behavior of curvature discontinuities at boundaries.

A. Validity regime of the EFT

The EFT is valid when all curvature invariants and field gradients remain below the ultraviolet cutoff scale Λ . Specifically, we require

$$|\partial\phi| \ll \Lambda^2, \quad |\nabla^2\phi| \ll \Lambda^3, \quad |k_{\text{loc}}| \ll \Lambda^2. \quad (19)$$

The core radius must satisfy

$$r_c \gg \Lambda^{-1}, \quad (20)$$

ensuring that the localized configuration is well within the infrared regime.

Higher-derivative operators such as $(\nabla^2\phi)^2$ may appear in the EFT Lagrangian, but their coefficients must be suppressed by powers of Λ to avoid exciting additional propagating degrees of freedom (e.g., Ostrogradsky ghosts). We assume throughout that the EFT coefficients satisfy the standard positivity and analyticity constraints of low-energy effective theories.

B. Gradient extremum and core stationarity

A physically admissible localized configuration must satisfy the Euler–Lagrange equation $\delta E/\delta\rho = 0$ derived from the static energy functional

$$E[\rho] = 4\pi \int_0^\infty dr r^2 \mathcal{E}[\rho(r), \partial_r \rho(r), \dots], \quad (21)$$

where $\mathcal{E}$ is the local energy density.

In addition to regularity and normalizability, we impose the *core stationarity condition*

$$\left. \frac{d}{dr} |\partial_r \rho(r)| \right|_{r=r_c} = 0, \quad (22)$$

which ensures that the core radius r_c corresponds to an extremum of the radial gradient. This condition, together with the curvature activation normalization $N_K = 1$, uniquely fixes r_c for a given profile family.

C. Stability analysis of localized modes

To analyze stability, consider small perturbations $\delta\rho(r)$ around a static localized solution $\rho_\star(r)$:

$$\rho(r) = \rho_\star(r) + \delta\rho(r). \quad (23)$$

Expanding the energy functional to quadratic order yields

$$E[\rho_\star + \delta\rho] = E[\rho_\star] + \frac{1}{2} \int_0^\infty dr \delta\rho(r) \mathcal{H}[\rho_\star] \delta\rho(r) + \mathcal{O}(\delta\rho^3), \quad (24)$$

where $\mathcal{H}$ is a self-adjoint stability operator.

In the regime where gradient and curvature terms dominate, $\mathcal{H}$ takes the schematic form

$$\mathcal{H}[\rho_\star] \simeq -A(r) \frac{d^2}{dr^2} - B(r) \frac{d}{dr} + C(r; k_{\text{loc}}), \quad (25)$$

with $A(r) > 0$ and $C(r; k_{\text{loc}})$ increasing with k_{loc} near r_c .

The sign of the lowest eigenvalue of $\mathcal{H}$ determines stability:

- $k_{\text{loc}} > 0$: The curvature contribution $C(r; k_{\text{loc}})$ stabilizes the configuration. The lowest eigenvalue is non-negative, and the mode is linearly stable.
- $k_{\text{loc}} = 0$: The curvature contribution vanishes. Only marginally bound or delocalized modes exist.
- $k_{\text{loc}} < 0$: The curvature contribution lowers $C(r; k_{\text{loc}})$, making negative eigenvalues likely. No static, stable localized solutions exist.

Thus, the curvature threshold $k_{\text{loc}} > 0$ is a necessary condition for the existence of stable localized modes.

D. Curvature discontinuities and shockfronts

Localized configurations may exhibit curvature discontinuities at boundaries, particularly when coupled to external fields or conductive shells. The jump in the extrinsic curvature tensor across a boundary is given schematically by

$$\Delta K_{ij} \sim \frac{T_{ij}}{\rho_c e^{-\kappa_n r}}, \quad (26)$$

where T_{ij} is the stress–energy tensor of the external field, ρ_c is the central density, and κ_n is a decay parameter controlling the penetration of curvature into the core.

These curvature discontinuities propagate as *shockfronts* in the scalar topographic field and play a central role in the experimental proposals of Sec. VI.

E. Positivity, unitarity, and causality

The EFT coefficients entering $\mathcal{L}_{\text{curv}}$ must satisfy the standard positivity and analyticity constraints of low-energy effective field theory. In particular:

- Higher-derivative operators must be suppressed by powers of Λ to avoid propagating ghostlike degrees of freedom.
- The effective metric for fluctuations must remain Lorentzian.
- Characteristic cones must lie within or coincide with the light cone defined by $g_{\mu\nu}$.

Throughout this work we assume that $\mathcal{L}_{\text{curv}}$ is chosen such that these conditions hold, allowing k_{loc} to be treated as an emergent geometric parameter rather than a dynamical ghostlike degree of freedom.

This completes the analysis of EFT validity and stability.

V. PROFILE INDEPENDENCE AND REPRESENTATIVE ANSATZ

The mass functional introduced in Sec. III depends only on the effective local curvature scalar k_{loc} and the core radius r_c . In this section we show that these quantities are insensitive to the detailed shape of the scalar density profile $\rho(r)$, provided certain continuity and stationarity conditions are satisfied. We then introduce representative Gaussian and hybrid profiles to illustrate the general structure.

A. Profile independence

Let $\rho_1(r)$ and $\rho_2(r)$ be two localized, spherically symmetric profiles satisfying:

(a) continuity at the core radius,

$$\rho_1(r_c) = \rho_2(r_c), \quad (27)$$

(b) gradient extremum at the core,

$$\left. \frac{d}{dr} |\partial_r \rho_1(r)| \right|_{r=r_c} = \left. \frac{d}{dr} |\partial_r \rho_2(r)| \right|_{r=r_c} = 0, \quad (28)$$

(c) identical curvature activation number,

$$N_{\mathcal{K}}[\rho_1] = N_{\mathcal{K}}[\rho_2] = 1. \quad (29)$$

Under these conditions, the curvature operators satisfy

$$\mathcal{K}[\rho_1](r_c) = \mathcal{K}[\rho_2](r_c), \quad (30)$$

and therefore

$$k_{\text{loc}}[\rho_1] = k_{\text{loc}}[\rho_2]. \quad (31)$$

Thus, the mass functional

$$m = \frac{c^2}{2G} r_c^2 k_{\text{loc}} \quad (32)$$

is *profile independent*. The detailed shape of $\rho(r)$ does not affect the assigned mass as long as the curvature activation is preserved.

B. Gaussian core ansatz

A convenient representative profile is the Gaussian core,

$$\rho_G(r) = \rho_c \exp\left[-\frac{(r - r_c)^2}{2\sigma^2}\right], \quad (33)$$

where ρ_c is the central density and σ is the width.

For this profile,

$$\ln \rho_G(r) = \ln \rho_c - \frac{(r - r_c)^2}{2\sigma^2}, \quad (34)$$

and the curvature operator becomes

$$\mathcal{K}[\rho_G](r) = \frac{1}{\sigma^2} - \frac{(r - r_c)}{\sigma^2 r} - \frac{(r - r_c)^2}{\sigma^4}. \quad (35)$$

Evaluating at $r = r_c$ yields

$$k_{\text{loc}}^{(\text{G})} = \frac{1}{\sigma^2}. \quad (36)$$

The activation number becomes

$$N_{\mathcal{K}}^{(\text{G})} = \int_{r_c - \sigma}^{r_c + \sigma} \sigma \mathcal{K}[\rho_G](r) dr = 1, \quad (37)$$

fixing the normalization of σ for a single activation.

C. Hybrid Gaussian–exponential profile

To model configurations with a sharp core and exponential halo, we introduce the hybrid profile

$$\rho_H(r) = \begin{cases} \rho_c \exp\left[-\frac{(r - r_c)^2}{2\sigma^2}\right], & r < r_c, \\ \rho_c \exp\left[-\frac{(r - r_c)}{L}\right], & r \geq r_c, \end{cases} \quad (38)$$

where L is the exponential decay length.

Continuity at r_c is automatic. The gradient extremum condition (28) fixes a relation between σ and L :

$$\partial_r \rho_H(r)|_{r=r_c^-} = \partial_r \rho_H(r)|_{r=r_c^+} \Rightarrow \frac{r_c - r_c}{\sigma^2} = \frac{1}{L}, \quad (39)$$

yielding

$$L = \sigma. \quad (40)$$

Evaluating the curvature operator at r_c gives

$$k_{\text{loc}}^{(\text{H})} = \frac{1}{\sigma^2}, \quad (41)$$

identical to the Gaussian case.

Thus, the hybrid profile yields the same mass as the Gaussian profile, confirming profile independence.

D. Gaussian integrals

The normalization and curvature activation conditions rely on standard Gaussian integrals:

$$\int_{-\infty}^{\infty} e^{-\frac{(r-r_c)^2}{2\sigma^2}} dr = \sqrt{2\pi} \sigma, \quad (42)$$

$$\int_{-\infty}^{\infty} r^2 e^{-\frac{(r-r_c)^2}{2\sigma^2}} dr = \sqrt{2\pi} \sigma^3 (r_c^2 + \sigma^2). \quad (43)$$

These relations ensure that the curvature activation number N_K can be normalized consistently for any smooth profile family.

E. Basis expansion and functional invariance

More generally, let

$$\rho(r) = \sum_{k=1}^K \alpha_k \phi_k(r), \quad (44)$$

where $\{\phi_k(r)\}$ is any complete basis of localized functions and α_k are coefficients.

If the conditions (27)–(29) are satisfied, then

$$k_{\text{loc}}[\rho] = k_{\text{loc}}[\rho_G], \quad (45)$$

and the mass functional is invariant under basis transformations.

This completes the demonstration of profile independence.

VI. ELECTROMAGNETIC COUPLING AND EXPERIMENTAL TESTBEDS

The scalar topographic EFT admits a minimal, gauge-invariant coupling between the scalar field and the electromagnetic field strength. This coupling allows electromagnetic boundary conditions to excite curvature in the scalar sector, leading to measurable mass shifts in localized modes. In this section we formalize the coupling, derive the induced curvature shift, and propose experimental tests capable of probing the mass functional introduced in Sec. III.

A. Minimal electromagnetic coupling

We introduce the interaction Lagrangian

$$\mathcal{L}_{\text{int}} = -\frac{\zeta}{4} \phi^2 F_{\mu\nu} F^{\mu\nu}, \quad (46)$$

where ζ is a dimensionless coupling constant and $F_{\mu\nu}$ is the electromagnetic field tensor. This term modifies the effective curvature operator $\mathcal{K}[\rho]$ in regions where electromagnetic fields are present.

The interaction (46) is the lowest-dimension gauge-invariant operator coupling ϕ to electromagnetism and is consistent with EFT power counting.

B. Boundary-induced curvature excitation

Conductive boundaries modify the electromagnetic mode structure and therefore the expectation value of $F_{\mu\nu} F^{\mu\nu}$. Near a conductive shell, the effective curvature scalar shifts by

$$\delta k_{\text{loc}} \sim \zeta \frac{\langle F_{\mu\nu} F^{\mu\nu} \rangle_{\text{shell}}}{\Lambda^2}, \quad (47)$$

where Λ is the EFT cutoff scale.

For configurations dominated by electric fields,

$$\langle F_{\mu\nu} F^{\mu\nu} \rangle_{\text{shell}} \sim E^2 \sim \left(\frac{V_{\text{drive}}}{R} \right)^2, \quad (48)$$

where V_{drive} is the applied RF voltage and R is a characteristic geometric scale (e.g., the radius of a conductive ring).

C. Mass shift from electromagnetic drive

For a localized scalar mode with core radius comparable to the shell radius, the mass shift induced by curvature excitation is

$$\delta m \sim \frac{c^2}{2G} R^2 \delta k_{\text{loc}} \sim \frac{c^2}{2G} \zeta \frac{V_{\text{drive}}^2}{\Lambda^2} \mathcal{G}(f, d, R_{\text{rms}}), \quad (49)$$

where $\mathcal{G}$ is a dimensionless geometry/skin-depth factor depending on:

- the drive frequency f ,
- the conductive coating thickness d ,
- the surface roughness R_{rms} ,
- and the multilayer electromagnetic penetration structure.

The numerical prefactor

$$\frac{c^2}{2G} \approx 6.74 \times 10^{26} \text{ kg/m}, \quad (50)$$

sets the universal geometric scale of the mass shift.

D. Conservative torsion-balance interpretation

A torsion balance with minimal detectable force $F_{\text{min}} \sim 10^{-14}$ N corresponds to a detectable mass shift

$$\delta m_{\text{min}} \sim \frac{F_{\text{min}}}{g} \sim 10^{-15} \text{ kg}. \quad (51)$$

Using Eq. (49), this yields the detectability condition

$$\frac{\zeta}{\Lambda^2} \gtrsim 10^{-45}, \quad (52)$$

for representative values of V_{drive} and $\mathcal{G}$.

This bound is extremely weak, indicating that torsion balances are not optimal probes of the scalar topographic EFT.

E. Optimized mechanical resonator testbed

A far more sensitive approach uses a high- Q mechanical resonator with effective mass $m_{\text{eff}} \sim 10^{-12}$ – 10^{-9} kg and resonant frequency $\omega_m/2\pi \sim 10$ – 100 kHz. A conductive ring patterned on the resonator is driven at GHz frequencies, inducing a curvature-dependent mass shift.

The resonance frequency shifts according to

$$\frac{\delta \omega_m}{\omega_m} \approx -\frac{1}{2} \frac{\delta m}{m_{\text{eff}}}. \quad (53)$$

Modern optomechanical readout techniques achieve sensitivities

$$\frac{\delta\omega_m}{\omega_m} \sim 10^{-9} - 10^{-12}, \quad (54)$$

corresponding to detectable mass shifts as small as

$$\delta m_{\min} \sim 10^{-21} \text{ kg}, \quad (55)$$

six orders of magnitude better than torsion balances.

F. Drive-synchronous detection strategy

To isolate the curvature-induced mass shift, we modulate the RF drive slowly:

$$V_{\text{drive}}(t) = V_0 + \Delta V \cos(\Omega t), \quad \Omega \ll \omega_m. \quad (56)$$

Equation (49) implies

$$\delta m(t) \propto V_{\text{drive}}^2(t) = V_0^2 + 2V_0\Delta V \cos(\Omega t) + \mathcal{O}(\Delta V^2). \quad (57)$$

Thus, the mechanical frequency exhibits a drive-synchronous modulation

$$\delta\omega_m(t) \propto \cos(\Omega t), \quad (58)$$

which can be extracted using lock-in techniques.

This signature is difficult to mimic with electromagnetic forces, which act at the GHz drive frequency rather than the slow modulation frequency Ω .

G. Summary of experimental prospects

Electromagnetic curvature excitation provides a concrete, falsifiable pathway to testing the scalar topographic EFT. Mechanical resonators offer the most promising sensitivity, with the potential to probe mass shifts many orders of magnitude below current gravitational experiments.

A null result would place strong constraints on the coupling ratio ζ/Λ^2 , while a positive detection would provide direct evidence for curvature-induced mass generation.

VII. DISCUSSION

The scalar topographic effective field theory developed in this work provides a geometric mechanism for mass generation based on local curvature thresholds. The mass functional

$$m = \frac{c^2}{2G} r_c^2 k_{\text{loc}}, \quad (59)$$

is defined by a matching ansatz that ties the curvature activation of a localized scalar density profile to the infrared Newtonian potential. This construction is independent of the microphysical origin of the scalar field and relies only on geometric and EFT principles.

The framework exhibits several notable features:

- **Universality.** The mass depends only on the curvature activation number $N_{\mathcal{K}}$ and the effective curvature scalar k_{loc} , not on the detailed shape of the profile.
- **Stability threshold.** Stable localized modes require $k_{\text{loc}} > 0$, providing a geometric criterion for the existence of mass-bearing configurations.
- **EFT consistency.** The theory respects standard EFT bounds, including positivity, unitarity, and causality, provided higher-derivative operators are suppressed by the cutoff scale Λ .
- **Experimental accessibility.** Electromagnetic curvature excitation provides a concrete pathway to testing the theory, with mechanical resonators offering sensitivity to mass shifts as small as 10^{-21} kg.

The theory is intentionally minimal: it does not specify the microphysical origin of the scalar field, nor does it embed the curvature operator in a full covariant action. These extensions are left for future work.

VIII. CONCLUSION

We have presented a scalar topographic effective field theory in which mass arises from local curvature thresholds encoded in the curvature operator $\mathcal{K}[\rho] = -\nabla^2 \ln \rho$. The effective curvature scalar k_{loc} and the activation number $N_{\mathcal{K}}$ provide a geometric characterization of localized modes, and the mass functional

$$m = \frac{c^2}{2G} r_c^2 k_{\text{loc}} \quad (60)$$

ensures consistency with the Newtonian potential in the infrared.

We analyzed the EFT validity regime, stability conditions, and profile independence of the mass functional. We also proposed experimental tests based on electromagnetic curvature excitation in conductive shells, identifying mechanical resonators as the most promising platform for detecting curvature-induced mass shifts.

The framework is falsifiable: a null result in sufficiently sensitive resonator experiments would place strong constraints on the coupling ratio ζ/Λ^2 , while a positive detection would provide direct evidence for curvature-induced mass generation.

Future work includes embedding the curvature operator in a covariant action, exploring multi-activation configurations, and investigating potential connections to microphysical mass generation mechanisms.

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